

Week 3: Statistics & Probability

Statistical Analysis Summary Report

Datasets: Housing Prices & Titanic Survival

AnalystLab Africa Data Science Internship Program

Tools used: Python · Pandas · NumPy · SciPy · Seaborn · Matplotlib · Scikit-learn

1. Introduction.....	3
2. Part A — Housing Dataset.....	3
2.1 Descriptive Statistics.....	3
2.2 Probability Distributions.....	4
2.3 Hypothesis Testing.....	4
2.4 Correlation vs Causation.....	5
3. Part B — Titanic Dataset	7
3.1 Descriptive Statistics.....	7
3.2 Probability Distributions	7
3.3 Hypothesis Testing.....	8
3.4 Correlation vs Causation.....	9
4. Overall Key Insights	10
5. Conclusion	10

1. Introduction

This report presents the statistical analysis carried out in Week 3 of the Analyst Lab Africa Data Science Internship Program. Building on the data cleaning and exploratory data analysis completed in Weeks 1 and 2, this week focused on statistical thinking — the foundation underlying every data science model.

Two cleaned datasets were used for this analysis: the Housing Prices dataset and the Titanic Survival dataset. For each dataset, four areas of statistical thinking were explored: descriptive statistics, probability distributions, hypothesis testing, and the distinction between correlation and causation.

All analysis was performed using Python, with Pandas and NumPy for data handling, SciPy for statistical testing, and Seaborn/Matplotlib for visualization.

2. Part A — Housing Dataset

2.1 Descriptive Statistics

Descriptive statistics summarize the central tendency, spread, and shape of each numeric feature in the dataset.

Central Tendency

Column	Mean	Median	Mode
price	4,766,729	4,340,000	3,500,000
area	5,151	4,600	6,000
bedrooms	2.97	3	3
bathrooms	1.29	1	1
stories	1.81	2	2

The mean price (4.77M) is noticeably higher than the median price (4.34M), and higher still than the mode (3.5M). This gap is a strong early indicator of right skew — a small number of high-value luxury houses are pulling the average upward, while the typical (most common) house is considerably cheaper.

Spread

Column	Std. Deviation	Variance	Range
price	1,870,440	3.50×10^{12}	11,550,000
area	2,170	4.71×10^6	14,550
bedrooms	0.74	0.54	5
bathrooms	0.50	0.25	3

Price has by far the highest variance and standard deviation of all features, confirming it is the most spread-out variable in the dataset — consistent with it being measured in millions while other features are measured in small whole numbers.

Skewness & Kurtosis

Column	Skewness	Interpretation
price	1.21	Highly right-skewed
area	1.32	Highly right-skewed
hotwaterheating	4.35	Extremely skewed (rare feature)
bedrooms	0.50	Approximately normal
mainroad	-2.07	Left-skewed (majority = yes)

A skewness value between -0.5 and 0.5 is considered approximately normal; values above 1.0 indicate strong skew. Price, area, and bathrooms are all notably right-skewed, while hotwaterheating is extremely skewed because the feature is rare (present in only about 5% of houses).

2.2 Probability Distributions

To formally test whether price and area follow a normal distribution, three checks were applied: a histogram overlaid with a normal curve, the Shapiro-Wilk statistical test, and a Q-Q plot.

Column	Shapiro-Wilk p-value	Normally Distributed?
price	0.0000	No
area	0.0000	No
bedrooms	0.0000	No
bathrooms	0.0000	No

All four columns returned a p-value far below the 0.05 threshold, confirming statistically that none of them follow a normal distribution. The Q-Q plots supported this — points followed the reference line in the middle of the distribution but curved away sharply at both tails, which is the signature of a right-skewed distribution with outliers.

Applying a log transformation to price and area produced visibly more symmetric, bell-shaped histograms. This step is recommended before feeding these features into a linear model, as it reduces the influence of extreme values and better satisfies the normality assumptions behind linear regression.

2.3 Hypothesis Testing

Hypothesis tests were used to check whether observed differences in price across groups were statistically real or simply due to chance. The decision rule applied throughout: if the p-value is below 0.05, the null hypothesis (H_0 : no difference) is rejected in favour of a real effect.

Test	Comparison	p-value	Result
------	------------	---------	--------

T-test	Airconditioning vs Price	< 0.05	Significant effect
T-test	Mainroad vs Price	< 0.05	Significant effect
ANOVA	Furnishing Status vs Price	< 0.05	Significant effect
ANOVA	Bedrooms vs Price	< 0.05	Significant effect
T-test	Basement vs Price	< 0.05	Significant effect
Chi-square	Mainroad vs Airconditioning	0.0199	Related
Chi-square	Prefarea vs Furnishing Status	0.0381	Related

Every feature tested showed a statistically significant relationship with price, including features with comparatively weak correlation values, such as basement (correlation = 0.19). This highlights an important statistical distinction: a p-value answers whether an effect is real, while a correlation coefficient answers how strong that effect is. With a sample size of 545 houses, even modest real effects become statistically detectable.

The two chi-square tests showed that mainroad location is related to airconditioning ownership, and that preferred-area status is related to furnishing status — meaning these categorical features are not independent of one another in this dataset.

2.4 Correlation vs Causation

Correlation measures how strongly two variables move together; it does not establish that one causes the other. This distinction was tested directly using multiple linear regression and a confounding-variable check.

Correlation with Price (ranked)

Feature	Correlation
area	0.54
bathrooms	0.52
airconditioning	0.45
stories	0.42
parking	0.38
bedrooms	0.37

Regression Coefficients (effect holding other features constant)

Feature	Coefficient (₦/unit)	Interpretation
area	346.08	Each extra sq.ft. adds ~346 to price
bathrooms	1,214,389.70	Each extra bathroom adds ~1.21M to price
airconditioning	969,365.12	Having AC adds ~969K to price

stories	452,355.97	Each extra storey adds ~452K to price
---------	------------	---------------------------------------

Although area has the highest simple correlation with price (0.54), its coefficient shows a modest per-unit dollar effect (346 per sq.ft.). Bathrooms, with a slightly lower correlation (0.52), has by far the largest isolated effect on price once other features are held constant. This demonstrates that correlation strength and practical / causal impact are not the same thing.

A confounding-variable check was also run by comparing the area distribution of houses with and without airconditioning. Houses with AC had a noticeably higher median area (~6,000 sq.ft.) than houses without AC (~4,000 sq.ft.). This suggests that part of the apparent relationship between airconditioning and price may actually be driven by house size — AC may simply be a marker of larger, higher-quality homes rather than a direct cause of higher prices.

Conclusion: the housing dataset shows several strong correlations with price, but the data alone cannot prove which features directly cause price increases. Bathrooms and area remain the most reliable predictors, while features like airconditioning should be interpreted alongside potential confounding variables such as house size.

3. Part B — Titanic Dataset

3.1 Descriptive Statistics

The Titanic dataset (891 passengers) was analyzed after cleaning: Name, Ticket, Cabin, Unnamed: 0, and PassengerId were dropped, missing Age values were imputed, and Sex/Embarked were encoded numerically.

Statistic	Survived	Age	Fare
Mean	0.38	29.54	32.20
Median	0	29.0	14.45
Mode	0	29.0	8.05
Std. Dev.	0.49	13.01	49.69
Variance	0.24	169.36	2,469.44
Range	1	80	512.33

The mean of Survived (0.38) confirms that 38% of passengers survived. Fare shows the largest gap between mean (32.20) and median (14.45) of any feature analyzed across both datasets, and has by far the highest variance (2,469.44) and standard deviation (49.69) — confirming it is the most spread-out, unequal variable, reflecting the wealth gap among passengers.

Skewness & Kurtosis

Column	Skewness	Kurtosis	Interpretation
Fare	4.79	33.40	Extremely right-skewed and peaked
SibSp	3.70	17.88	Highly right-skewed
Parch	2.75	9.78	Highly right-skewed
Embarked	1.54	1.11	Right-skewed
Age	0.46	0.99	Approximately normal
Pclass	-0.63	-1.28	Slightly left-skewed, flat

Fare is the most extreme feature in either dataset, with a skewness of 4.79 and kurtosis of 33.40 — both far beyond the normal range. Age is the most well-behaved numeric feature, with skewness of 0.46, within the approximately-normal band of -0.5 to 0.5.

3.2 Probability Distributions

Fare and Age were tested for normality using a histogram with an overlaid normal curve, the Shapiro-Wilk test, and a Q-Q plot.

Column	Shapiro-Wilk p-value	Normally Distributed?
--------	----------------------	-----------------------

Fare	0.0000	No
Age	0.0000	No
SibSp	0.0000	No
Parch	0.0000	No

All four columns returned a p-value of 0.0000, failing the normality test. Notably, even Age — which visually looked closest to a bell curve and had the best-behaved Q-Q plot of any feature tested across both datasets — still fails the formal Shapiro-Wilk test, showing that visual closeness to normal is not the same as passing a statistical test. The Q-Q plot for Fare showed the most severe deviation seen in either dataset: points sat flat at zero, curved well below the reference line through the middle, then jumped sharply away at the top, with three outlier points sitting at the maximum fare of 512.33.

A log transformation (using $\log(\text{Fare} + 1)$ to safely handle the passengers who paid a fare of exactly 0) substantially compressed the extreme high-fare outliers and produced a noticeably more symmetric distribution, though still not perfectly normal. The Age distribution showed a sharp artificial spike at the imputed median value, a known side effect of simple median imputation that slightly distorts the true shape of the distribution.

3.3 Hypothesis Testing

Hypothesis testing on the Titanic dataset focused on validating the bivariate patterns observed during EDA — specifically, whether sex, class, fare, age, and embarkation port had a statistically real relationship with survival.

Test	Comparison	p-value	Result
Chi-square	Sex vs Survived	< 0.05	Significant relationship
Chi-square	Pclass vs Survived	< 0.05	Significant relationship
T-test	Fare: Survived vs Died	< 0.05	Significant difference
T-test	Age: Survived vs Died	< 0.05	Significant difference
ANOVA	Fare across Embarked ports	0.0000	Significant difference

Chi-square tests were used for Sex and Pclass since both are categorical variables being compared against the categorical Survived outcome. T-tests compared the continuous Fare and Age variables between the two survival groups, while ANOVA was used to compare Fare across all three embarkation ports at once, since that comparison involves more than two groups.

Every test rejected the null hypothesis, confirming that the differences seen in the bivariate EDA charts were statistically real rather than due to chance. This is consistent with the Titanic disaster being one of the most well-documented historical events in applied statistics: the survival advantages tied to sex, class, fare, and embarkation port were strong and consistent enough to register clearly even in a sample of under 900 passengers. As with the Housing dataset, a significant p-value confirms an effect is real, but the practical size of the effect — such as the 75% vs 19% survival gap between sexes — is what makes a finding meaningful, not the p-value alone.

3.4 Correlation vs Causation

As with the Housing data, correlation between features and survival does not by itself prove causation. A multiple logistic regression was used to isolate each feature's effect on survival while holding the others constant, and a boxplot of Fare by Pclass was used to test directly for confounding.

Correlation with Survived (ranked)

Feature	Correlation
Sex	0.54
Fare	0.26
Embarked	0.11
Parch	0.08
SibSp	-0.04
Age	-0.07
Pclass	-0.34

Logistic Regression Coefficients (effect holding other features constant)

Feature	Coefficient	Interpretation
Sex	+2.5132	Being female sharply increases survival odds, even controlling for class, age and fare
Pclass	-1.1143	Each step toward a lower (worse) class sharply decreases survival odds
Age	-0.0326	Slight decrease in survival odds with each additional year
Fare	+0.0008	Effectively zero effect once class is controlled for

This is the key causation finding of the analysis. In the simple correlation, Fare appears moderately related to survival (0.26). However, a boxplot of Fare grouped by Pclass shows that first-class passengers paid dramatically more than second- or third-class passengers (median fare roughly 60 in 1st class versus 10–15 in 3rd class), and Fare correlates strongly with Pclass itself (-0.55). Once Pclass is included in the regression, Fare's coefficient collapses to virtually nothing (0.0008), while Pclass remains a strong, significant predictor (-1.1143).

This demonstrates that Fare was a confounded variable: its apparent relationship with survival was being driven by its strong association with Pclass, not by ticket price itself influencing survival outcomes. Passenger class — more plausibly tied to cabin location and lifeboat proximity — is the more fundamental factor.

Sex remains the strongest predictor in the regression by a wide margin (+2.5132) and is the feature in this dataset that comes closest to a true causal relationship, since the “women and children first” evacuation protocol was an explicit, documented policy rather than an indirect statistical association.

Conclusion: survival on the Titanic was most strongly associated with sex and class. Fare's raw correlation with survival is misleading on its own — it is almost entirely explained by class once a multiple regression controls for it, a clear real-world example of correlation without causation.

4. Overall Key Insights

- Both datasets contain features that are statistically significant predictors of their target variable, but statistical significance (p-value) and practical strength (correlation) are not the same thing — features can be “real” but weak, or strongly correlated but confounded by another variable.
 - Neither dataset's key numeric features (price, area, fare, age) are normally distributed. Log transformation is recommended for price, area, and fare before they are used in linear models.
 - In the Housing dataset, bathrooms has the strongest isolated effect on price once other features are controlled for, even though area has a higher simple correlation — area's effect on price may be partly confounded by other factors.
 - In the Titanic dataset, Fare's correlation with survival (0.26) is almost entirely explained by Passenger Class once a multiple logistic regression controls for it (Fare's coefficient drops to 0.0008) — a clear, data-backed example of correlation without causation.
 - Sex is the strongest and most direct predictor of survival on the Titanic, closely tied to the historically documented “women and children first” evacuation policy.
 - Across both datasets, simple imputation methods (median/mode) are convenient but can distort the true distribution of a feature — more advanced methods such as KNN imputation are recommended for production-quality analysis.
-

5. Conclusion

Week 3 applied core statistical thinking — descriptive statistics, probability distributions, hypothesis testing, and the correlation-versus-causation distinction — to the Housing and Titanic datasets that were cleaned and explored in Weeks 1 and 2. The analysis confirmed that both datasets are right-skewed and not normally distributed, identified the features most strongly and most reliably associated with the target variable in each case, and demonstrated concrete examples of confounding variables that complicate any claim of direct causation.

These statistical foundations will directly inform the modelling choices in the coming weeks: log transformations for skewed numeric features, careful feature selection based on both correlation and regression coefficients, and an awareness that strong predictors identified here (bathrooms for price, sex for survival) should be prioritized when building predictive models.